

2026-10-02_read_vdl-tools

QUESTION	What does the vDL already hold that CanVS can use, and what does it change
READ	Gasparrini, CTS_smallarea; Éric, Extract mortality 2010 to 2023, extract_estimates_RDC, CTS QUEBEC; Lulu, 1) Data Management and 2) Settings and Cross-Basis Specifications; the 09-28 Éric meeting; DAD Vetting Handbook (contents); Stat/Transfer User Manual (contents); CVSD RDC User Guide §5.4; VMA Researcher Manual; RDC / FRDC Confidentiality Handbook for Economic Microdata; RDC Confidentiality Vetting Request Form. Thirteen tracker rows (SOFTWARE), thirty-four shelf rows
FINDING	The 21-lag cross-basis needs no full DA $\times$ day grid; lags come from a date-offset join onto the rows that exist
MOVED	lag route, from grid rebuild to one join; outcome ruling, from re-extraction to rebuild; covariance gate, from banned to unsettled; mortality_cts, from filtered object to probable full grid
SHELVED	thirty-four rows: vDL and StatCan documentation, contracts and agreements, data sources, R packages, repositories, reference tables, the HC-10114 workspace tree

Finding

The 21-lag cross-basis needs no full DA $\times$ day grid. Éric and Lulu build lags the same way. Each row looks up its lag days in a complete exposure table, matched on DA and DATE - k, and off-series dates return NA and drop from the fit. Gasparrini does the opposite. He pads the grid with zero-death days and shifts within each area. The two routes give the same coefficients under the conditional likelihood, because strata with no deaths carry no information; only the quasi-Poisson dispersion differs. The filtered rows already inside the vDL are therefore enough. What the 21-lag matrix needs is a complete Daymet-by-DA table, and that table is already in Transfer/Incoming.

Per source

Gasparrini, CTS_smallarea. The reference code for the small-area case time series, London MSOAs over two summers. It completes the area $\times$ day grid with zeros, links gridded temperature by area-weighted extraction, and compares temporal against spatial variation. The model files 03.mainmod and 04.intmod were not read.

Éric, Extract mortality 2010 to 2023. The SAS program behind the project's mortality data. Per year it merges the health and geography files on registration_number, flags ten causes from the ICD code, drops the geography file's DA and relinks every postal code through one PCCF vintage. It then builds a time-stratified case-crossover file and, from its cases, a warm-season case time series.

Éric, extract_estimates_RDC. The SMOKE Study's estimate and release script. It fits smoke and non-smoke PM2.5 cross-bases with temperature and humidity adjusted, and writes every figure and table as CSV. It never writes coefficients or their covariance. Its comments carry the reasoning of

each version, the memory arithmetic of a cross-basis on tens of millions of rows, and the rule that rounding is for reading, not disclosure.

Éric, CTS QUEBEC. Practice code on Québec counts by RSS grouped into public health units. It runs the full two-stage chain: a gnm fit per unit, `crossreduce`, `mixmeta` with and without meta-predictors, a Wald test, BLUPs, the minimum-risk temperature per unit inside p25–p90, recentring and a pooled curve. Four bugs keep it from running as written.

Lulu, 1) Data Management and 2) Settings. Her pipeline for PM2.5 and heat. She reads `mortality_cts` from SAS once in about 2.5 hours and saves every later step as its own object. She builds $DA \times year \times month \times weekday$ strata, drops strata with no deaths, and joins DA-level PM2.5 by date offset. Her settings note that the lag window goes to 21 by rebuilding the lags in Step 1.

The 09-28 Éric meeting. Three facts. His VM and Lulu’s have about 256 GB against 16 for students. Every year is relinked through one PCCF vintage. `mortality_cts` is all-cause only.

DAD Vetting Handbook. The vetting rulebook for this file, version 2.0, May 2025. Only the contents were read. Models sit on p. 16, Graphs on p. 18, Heat Maps on p. 19 and Residuals With Geography on p. 42.

Stat/Transfer User Manual. The SAS-to-R converter in Researcher Resources. Only the contents were read. Selecting Cases, Selecting Variables and SAS Reading are the sections that could cut the 2.5-hour read.

CVSD RDC User Guide §5.4. One section. It names no vetting rule for the linked data and sends the reader to the RDC Analyst and the vetting training materials.

VMA Researcher Manual. How output leaves the vDL. Files are staged in the project’s Vetting Requests folder from the VM, and the request is filed in the VMA from a browser outside it. Statuses run Draft, Submitted, Under review, Changes requested, Approved or Not approved, and released files land in the project’s outgoing storage.

RDC / FRDC Confidentiality Handbook for Economic Microdata. Written for business data. It releases final outputs only, forbids suppression of cells and of map areas, and lists coefficient variance-covariance matrices among model outputs that can never be released. It allows a non-numerical summary of signs and significance before the final release.

RDC Confidentiality Vetting Request Form. The general form, revised May 2020. Its checklist asks how linked data were linked, whether a covariance matrix is requested and with what supporting counts, and whether the output is final.

Across rows

1. **Non-accidental counts are a rebuild, not a re-extraction.** `smoke.mortality_2010_2023` holds every death record with `death_cause_4digits`, and the `noext` flag already exists (001–799 or A00–R99). (*mortality_cts*, Éric’s extraction)
2. **The minimum-risk temperature needs a search window.** Éric reads it from the BLUPs inside p25–p90. On 09-30 the hot tail turned down past the p90 knot and pulled the minimum out of band. The plan says p1–p99. (Éric’s Québec example, Lulu’s Step 1 object)
3. **Row-indexing a basis does not transfer.** Éric’s `subrows()` slices one national basis

because his basis is data-independent, a linear exposure with fixed lag knots. Ours puts knots at each CMA's p10, p75 and p90. (*Éric's release script, Éric's Québec example*)

4. **No map area may be dropped.** Area suppression is not permitted; a DA that cannot be released is collapsed or redesigned, never blanked. (*RDC handbook, vetting request form*)
5. **Non-numerical check-ins are written policy.** Before the final release, signs and significance can be vetted in place of numbers. (*RDC handbook, VMA manual*)
6. **mortality_cts is probably already a full grid.** Lulu applies the zero-event filter herself after reading it, and the SAS step that builds `WORK.TSDSTIMESERIESDATA`, where zero days would be padded, is missing from the code. (*mortality_cts, Éric's extraction, Lulu's pipeline*)
7. **Temperature arrives with the deaths.** `T_MEAN` and its lags sit inside `mortality_cts`; Lulu adds only PM2.5, from `CanOSSEM`. (*mortality_cts, Lulu's pipeline, Lulu's Step 1 object*)

My model

The 21-lag matrix for Stage 1 is one join of Daymet-by-DA at `DATE - k, k in 0:21`, onto the filtered rows. The full-grid builder leaves the critical path.

The outcome ruling costs a rebuild from a file that already exists, whichever way it goes.

The minimum-risk temperature is searched inside a stated window. The window is a written choice in the spec.

Each CMA builds its own cross-basis on its own series.

The November map collapses any unreleasable DA into a releasable unit and drops none. The release carries surfaces, never coefficients or covariance, as one final request.

Contradictions

1. **Cause.** Éric says `mortality_cts` is all-cause only. Lulu's Step 1 object carries cause-specific columns. Settled by `names(full)` and her unread `Heat Only - Data Management.R`.
2. **Covariance.** The general request form asks only for supporting counts when a covariance matrix is requested. The economic handbook bans coefficient covariance outright. CVSD §5.4 names no rule. Settled by the DAD Vetting Handbook, pp. 16, 18 and 19.
3. **Age bands.** Lulu's object has five, the blueprint has four. Settled by the band edges in her 1) `Data Management.R`.

Corrections

1. Lulu was thought to have joined temperature at health-region level. `T_MEAN` and its lags arrive inside `mortality_cts`. (*Lulu's pipeline*)

2. The column positions 14:21 and 22:29 were read as belonging to `full`. They belong to the sibling `Mortality Data_Step 1`. (*Lulu's Step 1 object*)
3. The 6 GB file that froze the VM was taken to be the Daymet file. Which file it was is not recorded. (*DA Daymet series*)
4. Éric's Québec example was read as a possible spec. It is practice code with four bugs: an undefined `mod`, NULL names from `cities$PHU`, an undefined `cpall`, and lag columns taken by position. (*Éric's Québec example*)

Open questions

1. Does `mortality_cts` pad zero-death days, or does it hold only days with a death?
2. At what resolution is the `T_MEAN` inside `mortality_cts`, DA or health region?
3. Which rule binds coefficient covariance for CVSD output, the general form's or the economic handbook's?
4. Does the rule against multiple iterations of output bind health projects as it binds economic ones?
5. Is a per-DA surface a map of model-predicted values in the handbook's sense, or an area-level summary outside that rule?
6. What window should the minimum-risk temperature search use for CanVS, and on what ground?

People

Éric Lavigne. Two registers. His production code is exacting: a pre-flight that stops on any missing column, mass-conservation checks on every partition, change logs that explain why each version broke and what the fix touches, and the reminder to pre-specify primary outcomes and state the test in Methods. His practice code is loose and fast, written to learn a method, bugs left in. He extracts in SAS and models in R. His release habit is the conservative one, curves and burden out, coefficients and covariance in.

Lulu. Writes to be read by a learner. Nearly every line carries a comment saying what it does, alternatives stay in the file commented out, and she flags what she does not understand rather than guessing (*has 98 more variables but unsure how these were calculated*). She saves every step as its own object, which is why her work fits a 16 GB machine. Her headers credit the lineage, *Modifies code that Eric created*.

Kafi. The curator. Placed Gasparrini's tutorial in the shared workspace verbatim, so the group works from the reference code, not a copy of a copy.

Antonio Gasparrini. The design's author, and his code is as spare as his papers. Capitalised comment headers, one idea per block, `data.table` and the native pipe, and order exploited deliberately (*important for keeping the time series sequence*). He keeps the code public and current on GitHub and says so in the first lines.

The DAD Vetting Committee. Risk-averse by charter. *If these goals conflict, and where judgement must be applied, reducing the risk of disclosure takes precedence.* Its documents prefer redesign to suppression, final output to iteration, and words to numbers when a number is not needed.